Unit 1.0 — Integral Calculus and its Applications

1.1 Concept of Integration

Integration is the reverse process of differentiation. If `d/dx [F(x)] = f(x)`, then the integral of `f(x)` with respect to `x` is `F(x) + C`, where C is the constant of integration: `∫ f(x) dx = F(x) + C`

1.2 Standard Integrals (Working Rules)

Function	Integral
`x^n`	`x^(n+1)/(n+1) + C` (n ≠ -1)
`1/x`	`ln|x| + C`
`e^x`	`e^x + C`
`sin x`	`-cos x + C`
`cos x`	`sin x + C`
$\sec^2 x$	`tan x + C`
$1/(1+x^2)$	$\tan^{-1}x + C$

1.3 Methods of Integration

Method of Substitution: Substitute `u = g(x)` to simplify the integral. **Example:** $\int 2x\cdot\cos(x^2)\,dx$. Let $u = x^2$, then `du = 2x dx`. $= \int \cos u\,du = \sin u + C = \sin(x^2) + C$

Trigonometric transformation: Using identities to simplify integrals involving trigonometric functions before integrating. **Example:** $\int \sin^2 x\,dx = \int (1 - \cos 2x)/2\,dx = x/2 - \sin 2x/4 + C$

Integration by parts: Used for the product of two functions: `∫ u·v dx = u∫v dx − ∫ [du/dx · ∫v dx] dx` **Example:** `∫ x·e^x dx = x·e^x − ∫ e^x dx = x·e^x − e^x + C`

Partial fractions: Used to integrate rational functions by splitting into simpler fractions. **Example:** `∫ 1/[(x-1)(x+2)] dx = (1/3)∫[1/(x-1) − 1/(x+2)] dx = (1/3) ln|(x-1)/(x+2)| + C`

1.4 Applications: Area and Volume

Area under a curve: The area bounded by curve `y = f(x)`, the x-axis, and the lines `x = a` and `x = b` is given by: $\text{Area} = \int_a^b f(x)\,dx$

Volume of solid of revolution: When the area is revolved about the x-axis: $V = \pi \int_a^b [f(x)]^2\,dx$

Example: Find the area between $y = x^2$ and the x-axis from `x=0` to `x=2`: $A = \int_0^2 x^2\,dx = [x^3/3]_0^2 = 8/3$ sq. units

Unit 2.0 — Number Theory

2.1 Euclid's Division Algorithm, GCD, LCM

Euclid's Division Lemma: For two positive integers a and b, there exist unique integers q and r such that $a = bq + r$, where $0 \le r < b$.

Euclid's Algorithm for GCD: Repeatedly apply the division lemma until the remainder is zero; the last non-zero divisor is the GCD. **Example:** Find GCD(48, 18): $48 = 18\times2 + 12$, $18 = 12\times1 + 6$, $12 = 6\times2 + 0$. GCD = 6.

Expressing GCD as a linear combination (ax+by): Using the Extended Euclidean Algorithm, GCD(a,b) can be written as $ax + by$ by back-substitution. **Example:** From above: $6 = 18 - 12\times1 = 18 - (48 - 18\times2) = 18\times3 - 48\times1$. So GCD(48,18) = 48(−1) + 18(3).

LCM: $\mathrm{LCM}(a,b) = (a \times b) / \mathrm{GCD}(a,b)$

2.2 Prime Numbers and Fundamental Theorem of Arithmetic

Prime number: A natural number greater than 1 having exactly two factors, 1 and itself.

Fundamental Theorem of Arithmetic: Every composite number can be expressed (factorized) as a product of primes, and this factorization is unique, apart from the order of factors. **Example:** $360 = 2^3 \times 3^2 \times 5$

2.3 Congruences and Modular Arithmetic

Congruence: Two integers a and b are said to be congruent modulo n, written $a \equiv b \pmod{n}$, if n divides $(a - b)$. **Example:** $17 \equiv 2 \pmod{5}$ since $17 - 2 = 15$ is divisible by 5.

Linear congruence: An equation of the form $ax \equiv b \pmod{n}$. It has a solution if and only if $\gcd(a,n)$ divides b. **Example:** Solve $3x \equiv 6 \pmod{9}$. Since gcd(3,9)=3 divides 6, solutions exist: $x \equiv 2 \pmod{3}$, giving $x = 2, 5, 8 \pmod{9}$.

Applications: Modular arithmetic forms the basis of cryptography (RSA algorithm), check-digit systems, and hashing techniques used in data security and compression.

2.4 Chinese Remainder Theorem (CRT)

The CRT states that if $n_1, n_2, \ldots, n_k$ are pairwise coprime, then the system of congruences: $x \equiv a_1 \pmod{n_1}$, $x \equiv a_2 \pmod{n_2}$, ... has a unique solution modulo $N = n_1\times n_2\times\ldots\times n_k$.

Example: Solve $x \equiv 2 \pmod{3}$ and $x \equiv 3 \pmod{5}$. $N = 15$. Testing values: $x = 8$ satisfies both ($8 \bmod 3 = 2$, $8 \bmod 5 = 3$). General solution: $x \equiv 8 \pmod{15}$.

Unit 3.0 — Numerical Solution of Nonlinear Equations

3.1 Algebraic and Transcendental Equations

Algebraic equation: An equation involving only algebraic terms (polynomials). Example: $x^3 - 4x + 1 = 0$

Transcendental equation: Contains transcendental functions (trigonometric, exponential, logarithmic). Example: $x - \cos x = 0$

3.2 Iterative Methods

Iterative methods find approximate roots by generating a sequence of successive approximations that converge towards the true root, continuing until the desired accuracy is achieved.

General fixed-point iteration: Rewrite $f(x) = 0$ as $x = g(x)$, then iterate: $x_{(n+1)} = g(x_n)$, starting from an initial guess, until $|x_{(n+1)} - x_n| < \varepsilon$ (desired accuracy).

3.3 Newton-Raphson Method

This method uses the tangent line approximation to iteratively converge to the root: $x_{(n+1)} = x_n - f(x_n)/f'(x_n)$

Example: Find a root of $f(x) = x^2 - 2 = 0$ near $x_0 = 1.5$. $f'(x) = 2x$ - $x_1 = 1.5 - (1.5^2 - 2)/(2\times1.5) = 1.5 - 0.25/3 = 1.4167$ - $x_2 = 1.4167 - (1.4167^2 - 2)/(2\times1.4167) \approx 1.41422$ The root converges to $\sqrt{2} \approx 1.41421$.

Engineering applications: Used to solve nonlinear equations arising in circuit design, structural analysis, control systems, and financial models (e.g., Black-Scholes model) where closed-form solutions are not available.

3.4 Bakhshali Iterative Method (IKS)

The **Bakhshali approximation**, from the ancient Indian Bakhshali manuscript, provides an iterative method to compute the approximate square root of a number N: $\sqrt{N} \approx a + d/2a - (d/2a)^2/[2(a + d/2a)]$ where a^2 is the nearest perfect square to N, and $d = N - a^2$.

Example: Approximate $\sqrt{41}$. Take $a = 6$ (since 6^2=36), $d = 41 - 36 = 5$. $\sqrt{41} \approx 6 + 5/12 - (5/12)^2/[2(6+5/12)] \approx 6.4031$ (actual value ≈ 6.4031) — showing remarkable accuracy.

Unit 4.0 — Linear Algebra

4.1 Rank of a Matrix

The **rank** of a matrix is the maximum number of linearly independent rows (or columns) in the matrix, found by reducing the matrix to row-echelon form and counting the number of non-zero rows.

Example: For `A = [[1,2,3],[2,4,6],[1,0,2]]`, R2 = 2×R1, so rank < 3. Reducing further gives rank(A) = 2.

4.2 Cayley-Hamilton Theorem

The **Cayley-Hamilton Theorem** states that every square matrix satisfies its own characteristic equation. If the characteristic equation of matrix A is $\lambda^n + c_1\lambda^{n-1} + ... + c_n = 0$, then $A^n + c_1A^{n-1} + ... + c_nI = 0$.

Use for finding inverse: From the characteristic equation, A^{-1} can be expressed as a polynomial in A, avoiding the need for cofactor expansion. **Example:** For a 2×2 matrix `A`, with characteristic equation $A^2 - (\text{trace } A)A + (\det A)I = 0$, we get: $A^{-1} = [(\text{trace } A)I - A] / \det(A)$

4.3 Eigenvalues and Eigenvectors

For a square matrix `A`, a non-zero vector `X` is an **eigenvector** with **eigenvalue** λ if: $AX = \lambda X$

Finding eigenvalues: Solve the characteristic equation $\det(A - \lambda I) = 0$.

Example: For `A = [[2,0],[0,3]]`: $\det(A-\lambda I) = (2-\lambda)(3-\lambda) = 0$ → eigenvalues $\lambda = 2, 3$. For $\lambda=2$: $(A-2I)X=0$ gives eigenvector $[1,0]^t$. For $\lambda=3$: eigenvector $[0,1]^t$.

Applications: Used extensively in Machine Learning for dimensionality reduction (PCA), feature extraction, and in vibration/structural analysis (natural frequencies).

4.4 Linearly Independent and Dependent Vectors

A set of vectors is **linearly independent** if no vector in the set can be written as a linear combination of the others (i.e., the only solution to $c_1v_1+c_2v_2+...=0$ is all $c_i=0$). A set is **linearly dependent** if at least one vector can be expressed as a combination of the others.

Example: $v_1=(1,2)$, $v_2=(2,4)$ are linearly dependent since $v_2 = 2v_1$. $v_1=(1,0)$, $v_2=(0,1)$ are linearly independent.

Unit 5.0 — Linear Programming

5.1 Mathematical Formulation of LPP

A **Linear Programming Problem (LPP)** involves optimizing (maximizing/minimizing) a linear objective function subject to a set of linear constraints and non-negativity restrictions.

General form:

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Maximize/Minimize: Z = c₁x₁ + c₂x₂ + ... + cₙxₙ
Subject to: a₁₁x₁ + a₁₂x₂ + ... ≤ (or ≥, =) b₁
            ...
            x₁, x₂, ... ≥ 0
```

Example formulation: A company makes products A and B. Product A needs 2 hrs machine time and gives profit ₹40; Product B needs 3 hrs and gives profit ₹50. Total machine time available = 60 hrs. `Maximize Z = 40x + 50y` `Subject to: 2x + 3y ≤ 60; x, y ≥ 0`

5.2 Graphical Method

Used for problems with **two variables**. Steps: 1. Plot the constraint lines on a graph. 2. Identify the feasible region (common region satisfying all constraints). 3. Find the corner points (vertices) of the feasible region. 4. Evaluate the objective function `Z` at each corner point. 5. Select the point giving the maximum (or minimum) value of Z.

Example: Maximize `Z = 3x + 5y` subject to `x + y ≤ 4`, `x ≥ 0, y ≥ 0`. Corner points: (0,0), (4,0), (0,4). Z values: 0, 12, 20. Maximum Z = 20 at (0,4).

5.3 Simplex Method

Used for problems with **more than two variables**, where the graphical method is not feasible. Steps: 1. Convert inequalities to equations by introducing slack/surplus variables. 2. Set up the initial simplex table (tableau) using the coefficients. 3. Identify the pivot column (most negative coefficient in the objective row, for maximization) and pivot row (minimum ratio test). 4. Perform row operations to update the tableau (pivot operation). 5. Repeat until no negative coefficients remain in the objective row — this gives the optimal solution.

Note: The Simplex method is widely applied in optimization problems for operations research, supply chain management, network flow problems, and Support Vector Machines (SVM) in machine learning.